Compte rendu

Routage VLAN / HSRP

Objectifs :

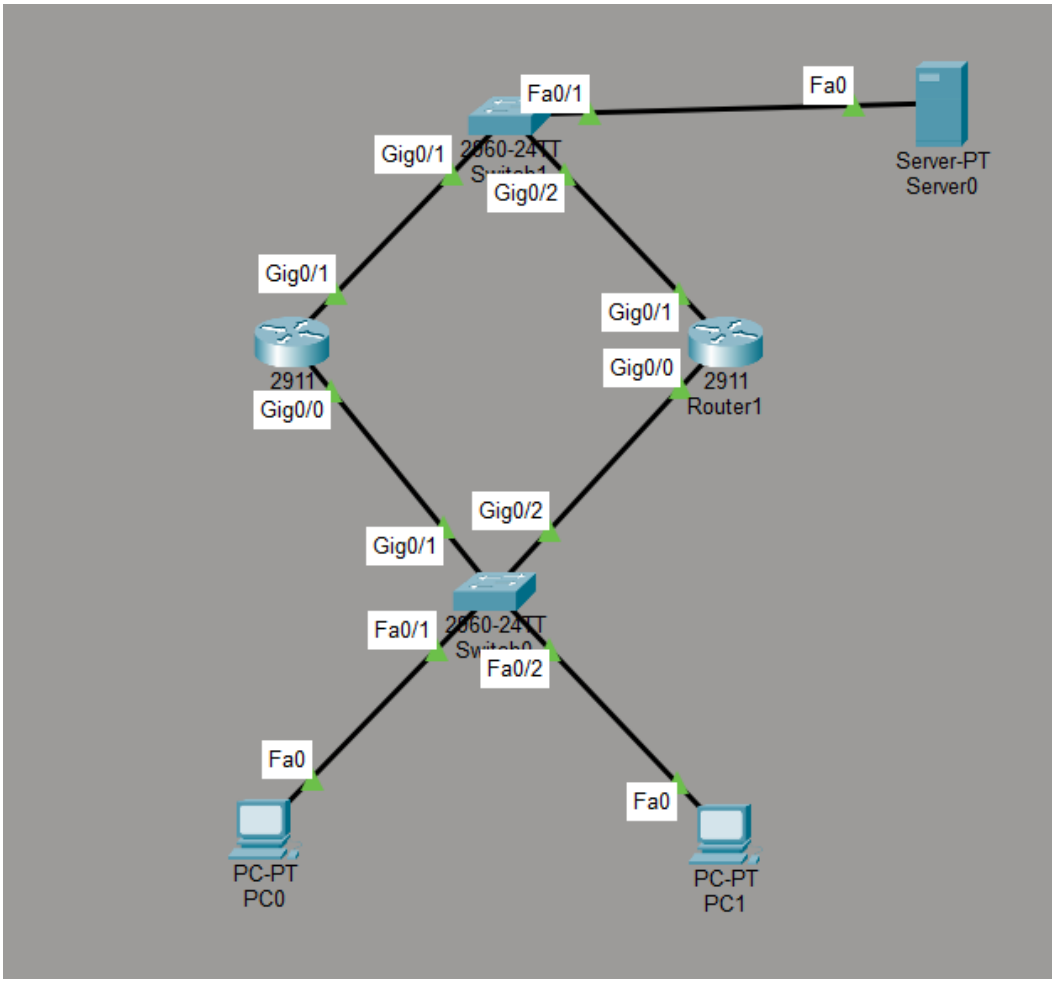
Mettre en place un reseau réseau qui permet :

Routage VLAN
Routeur actif = **R1 (vlan10)**
Routeur actif = **R2 (vlan20)**
HSRP avec une adresse VIP
Basculement automatique en cas de panne
Répartition de charge (load balancing) Haute disponibilité de la passerelle

Topologie

L’architecture comprend :

2 routeurs **R1** et **R2**
2 switches
VLAN 10 et VLAN 20 pour les utilisateurs
HSRP configuré sur chaque VLAN


Fa0/1
Fa0
Server-PT
Server0
Gig0/1
2960-24TT
Switch1
Gig0/2
Gig0/1
Gig0/1
2911
Gig0/0
Gig0/0
2911
Router1
Gig0/2
Gig0/1
Fa0/1
2960-24TT
Switch0
Fa0/2
Fa0
Fa0
PC-PT
PC0
PC-PT
PC1

Plan d’adressage

VLAN 10	VLAN 20
R1 = 192.168.10.1	R1 = 192.168.20.1
R2 = 192.168.10.2	R2 = 192.168.20.2
VIP = 192.168.10.254(passerelle des PC)	VIP = 192.168.20.254

Configuration du Switch Du Bas

```
vlan 10
name VLAN10
vlan 20
name VLAN20
interface fa0/23
switchport mode access
switchport access vlan 10
interface fa0/22
switchport mode access
switchport access vlan 20
interface g0/1
switchport mode trunk
interface g0/2
```

switchport mode trunk

Configuration du Switch du haut

interface g0/1
switchport mode access

Configuration R1 Router-on-a-Stick + HSRP

interface g0/0

no shutdown

interface g0/0.10

encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

standby 10 ip 192.168.10.254

standby 10 priority 120

standby 10 preempt

interface g0/0.20

encapsulation dot1Q 20

ip address 192.168.20.1 255.255.255.0

standby 20 ip 192.168.20.254

standby 20 priority 90

standby 20 preempt

interface g0/1
ip address 192.168.2.1 255.255.255.0

no shutdown

standby 30 ip 192.168.2.254

standby 30 priority 120

standby 30 preempt

Configuration R2 — Router-on-a-Stick + HSRP

interface g0/0
no shutdown
interface g0/0.10 encapsulation dot1Q 10
ip address 192.168.10.2 255.255.255.0
standby 10 priority 90
standby 10 preempt
standby 10 ip 192.168.10.254
interface g0/0.20

encapsulation dot1Q 20

ip address 192.168.20.2
255.255.255.0

standby 20 ip 192.168.20.254

standby 20 priority 120

standby 20 preempt

interface g0/1
ip address 192.168.2.2 255.255.255.0
no shutdown
standby 30 ip 192.168.2.254
standby 30 priority 90
standby 30 preempt

Répartition de charge

La priorité HSRP est configurée différemment :

Vlan10 = R1

Vlan20 = R2

Fonctionnement HSRP

HSRP crée une **passerelle virtuelle** partagée :

Les PC utilisent une IP virtuelle
Un routeur actif
L'autre est en standby
En cas de panne, basculement automatique

Procédure des tests

Vérification VLAN

```
Switch#
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/24
10   VLAN10                           active    Fa0/23
20   VLAN20                           active    Fa0/22
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Vérification trunk

```
Switch#sh
Switch#show inte
Switch#show interfaces tr
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1
Gig0/2      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005
Gig0/2      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20
Gig0/2      1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20
Gig0/2      1,10,20

Switch#
```

Sous-interfaces routeurs

R1 :

```
Router>en
Router#show ip inter
Router#show ip interface br
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

R2 :

```
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.3    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.2    YES manual up                    up
GigabitEthernet0/1     192.168.2.2     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

HSRP

R1 :

```
Router#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Gig0/1      30   120 P Active   local           192.168.2.2     192.168.2.254
Gig         10   120 P Active   local           192.168.10.3    192.168.10.254
Gig         20   90  P Standby  192.168.20.2    local           192.168.20.254
Router#
```

R2 :

```
Router#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Gig0/1      30   90  P Standby  192.168.2.1     local           192.168.2.254
Gig         10   100 P Standby  192.168.10.1    local           192.168.10.1
Gig         20   120 P Active   local           192.168.20.1    192.168.20.254
```

Ping depuis les pc

```
C:\>ping 192.168.10.254

Pinging 192.168.10.254 with 32 bytes of data:

Reply from 192.168.10.254: bytes=32 time<1ms TTL=255
Reply from 192.168.10.254: bytes=32 time<1ms TTL=255
Reply from 192.168.10.254: bytes=32 time=3ms TTL=255
Reply from 192.168.10.254: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.10.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

```
C:\>ping 192.168.20.254

Pinging 192.168.20.254 with 32 bytes of data:

Reply from 192.168.20.254: bytes=32 time<1ms TTL=255
Reply from 192.168.20.254: bytes=32 time<1ms TTL=255
Reply from 192.168.20.254: bytes=32 time=1ms TTL=255
Reply from 192.168.20.254: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.20.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Routage Inter-VLAN

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=10ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Vérification load balancing

VLAN10 vers Serveur :

```
C:\>tracert 192.168.2.10

Tracing route to 192.168.2.10 over a maximum of 30 hops:

  1   0 ms      1 ms      1 ms      192.168.10.1
  2   1 ms      0 ms      12 ms     192.168.2.10
```

PC VLAN20 vers Serveur :

```
C:\>tracert 192.168.2.10

Tracing route to 192.168.2.10 over a maximum of 30 hops:

  1   0 ms      1 ms      0 ms      192.168.20.2
  2   2 ms      0 ms      0 ms      192.168.2.10

Trace complete.
```

Basculement automatique

Simuler panne R2 :

interface g0/0 shutdown

Sur R1 :

show standby brief

```
Router#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Gig0/1      30   120 P Active   local           192.168.2.2     192.168.2.254
Gig         10   120 P Active   local           192.168.10.2    192.168.10.254
Gig         20   90  P Active   local           unknown         192.168.20.254
```

Les VLANS sont tout 2 actifs

Basculement validé PC0:

```
C:\>tracert 192.168.2.10

Tracing route to 192.168.2.10 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.10.1
  2   1 ms      1 ms      12 ms     192.168.2.10

Trace complete.
```

PC1:

```
C:\>tracert 192.168.2.10

Tracing route to 192.168.2.10 over a maximum of 30 hops:

  1   16 ms     1 ms      0 ms      192.168.20.1
  2   0 ms      0 ms      2 ms      192.168.2.10

Trace complete.
```